DOMAIN 3 STUDY QUESTIONS

Question 1

Question: In the process of risk assessment, risk can be classified into two general categories; societal risk and individual risk. Which of the following statements is the best description of a societal risk?

- a. The risk associated with a large group of individuals
- b. Number of incidences or consequences that can occur per year
- c. The risk of release of a hazardous chemical into the community
- d. All the risks that a given society is exposed to

Correct Answer: Select (a) **Explanation:** Societal risk is the risk associated with a large group of individuals.

Question 2

Question: Which of the following statements best describes the concept of individual risk?

- a. The risk associated with workplace hazards to an individual
- b. Number of undesirable incidences or consequences that can occur to an individual in any given day
- c. Probability of a single consequence occurring to an individual in a given year
- d. The risk an individual assumes in a given day

Correct Answer: Select (c) **Explanation:** Individual risk is the probability of a single consequence occurring to an individual in a given year.

Question 3

Question: Risk analysis in occupational and environmental safety:

- a. Is a qualitative method only
- b. Is a quantitative method only
- c. Can be qualitative or quantitative
- d. Must provide solutions to eliminate risks

Correct Answer: Select (c) **Explanation:** Risk analysis in occupational and environmental safety can be qualitative or quantitative.

Question 4

Question: In regard to the process of risk assessment, which of the following statements is the most correct?

- a. Risk must first be managed and then assessed.
- b. Risk can only be assessed after hazard identification.
- c. Risk must be assessed before hazard identification.
- d. Only (a) and (b) above.

Correct Answer: Select (b) **Explanation:** Risk assessment can only occur after hazard identification has taken place. Hazard identification is the first step in the risk management process.

Question 5

Question: MORT is used extensively in Total Quality Management (TQM). MORT is an acronym for:

- a. Management Oversight Risk Tree
- b. Minimum Operational Readiness Test
- c. Management Objectives and Reorganization Technique
- d. Management Organization and Restructure Technique

Correct Answer: Select (a) **Explanation:** MORT is an acronym for Management Oversight Risk Tree.

Question 6

Question: The Management Oversight Risk Tree (MORT):

- a. Starts with an undesirable event and traces that event to all its possible causes
- b. Is a diagram which primarily focuses on risk analysis and management failures to adequately respond to risks
- c. Is similar to fault tree analysis
- d. Only (a) and (c) above

Correct Answer: Select (d) **Explanation:** Management Oversight Risk Tree (MORT) starts with an undesirable event and traces that event to all its possible causes, and it is similar to fault tree analysis.

Question 7

Question: Unit Risk Exposure is calculated by:

- a. Dividing total lost dollars by the total number of units lost.
- b. Calculating the reliability of each component of a system and using the appropriate relationship for series or parallel system reliability.
- c. Dividing the total risk exposure by the total number of units.
- d. Calculating the reliability of each component within a system and adding the results for all components.

Correct Answer: Select (c) **Explanation:** Unit Risk Exposure is calculated by dividing the total risk exposure by the total number of units.

Question 8

Question: Risk identification must be conducted before:

- a. Risk management
- b. Risk analysis

- c. System safety application
- d. All of the above

Correct Answer: Select (d) **Explanation:** Risk identification must be conducted before risk management, risk analysis, and system safety applications. On complex, high-cost projects that are by nature uncertain, the risks can feed the rigorous process of assessment, analysis, mitigation and planning, allocation, and monitoring.

Question 9

Question: Proper risk assessment is one of the key steps in the process of hazardous waste management. Which of the following would you use as the first step in the process of risk assessment for a hazardous waste site?

- a. Site assessment
- b. Contact OSHA
- c. Contact the LEPC and SERC
- d. Contact the U.S. Coast Guard

Correct Answer: Select (a) **Explanation:** Proper site assessment provides information on the risks posed by a hazardous waste site.

The major objective of site assessment is to identify all conditions or suspected conditions that might be immediately dangerous to life or health. A secondary site assessment is conducted to identify these hazardous conditions and recommend appropriate protective measures. Examples of such hazards include, but are not limited to, entry into confined spaces, potentially explosive or flammable situations, visible vapor clouds, or areas where a biological indicator, such as dead animals or vegetation, is located.

Question 10

Question: In the process of risk assessment, both the probability and severity of an incident must be taken into account. From the following table, which case poses the highest risk?

Case	Probability	Severity
1	0.000005	2
2	0.000003	5
3	0.000001	6
4	0.000001	4

- a. Case 1
- b. Case 2
- c. Case 3
- d. Case 4

Correct Answer: Select (b) **Explanation:** One way to define risk is to multiply probability and severity. In this case:

- Case 1: 0.000010
- Case 2: 0.000015
- Case 3: 0.000006
- Case 4: 0.000004

Case 2 yields 0.000015, which is the highest risk value.

Question 11

Question: Which of the following statements is true in regards to cost-benefit analysis?

- a. Cost-benefit analysis calculates the benefits and costs of competing alternatives.
- b. The analysis can be performed by calculating the cost-to-benefit ratio.
- c. The analysis can be performed by calculating the benefit-to-cost ratio.
- d. All of the above.

Correct Answer: Select (d) **Explanation:** All statements are true in regards to cost-benefit analysis.

Cost-benefit analysis considers the equivalent dollar value of costs to benefits among competing projects. If the costs exceed the benefits the project is not considered worthwhile to pursue. Many components of costs and benefits lend themselves to simple analysis. However, there are other components for which intuition fails to offer an accurate method of measurement.

Question 12

Question: Hazard and Operability (HAZOP) studies are conducted to identify the deviations of process parameters from their design intent. A team conducts a HAZOP study. Which of the following IS NOT true in regards to HAZOP team leaders or their functions?

- a. The HAZOP team leader must be completely familiar with the process and its safety features.
- b. The HAZOP team leader must be able to assist in brainstorming the process.
- c. The HAZOP team leader must be able to document the findings of each brainstorming session.
- d. The HAZOP team leader must be able to prepare a sound HAZOP report.

Correct Answer: Select (a) **Explanation:** Although a HAZOP team leader must in general be familiar with the process, complete familiarity with the process is not a requirement. One of the responsibilities of the team leader is to form a multidisciplinary team that is completely familiar with the process and its safety features.

Question 13

Question: Which of the following is considered to be an environmental risk assessment technique?

- a. Probability analysis
- b. Systems analysis
- c. Cost-benefit analysis
- d. All of the above.

Correct Answer: Select (d) **Explanation:** All are techniques used in environmental risk assessment.

Question 14

Question: The risk assessment process must be conducted in the following order:

- a. Hazard accounting, hazard identification or risk evaluation, risk characterization
- b. Hazard identification, risk evaluation, risk characterization, hazard accounting
- c. Hazard identification, hazard accounting, risk characterization, risk evaluation
- d. Hazard identification, hazard accounting, risk evaluation, risk characterization

Correct Answer: Select (c) **Explanation:** Statement (c) describes the correct order for environmental risk assessment.

Question 15

Question: Residual risk is defined as equal to inherent risk less the effect of controls. Which of the following is NOT one of the three types of controls that are commonly used to reduce inherent risk?

- a. Preventive
- b. Probabilistic
- c. Detective
- d. Remedial

Correct Answer: Select (b) **Explanation:** Preventive controls focus on reducing the probability of risk occurring. Detective controls attempt to identify the occurrence of risk. If detective controls are applied before the occurrence of risk, they reduce the likelihood of occurrence. If they are applied after the occurrence of risk, they reduce the consequences of the risk. Remedial controls attempt to reduce the consequences of a risk that has occurred.

Question 16

Question: Which of the following statements is not true in regard to a management oversight risk tree (MORT)?

- a. It can be used to predict the adequacy of control elements already in place to prevent accidents.
- b. MORT uses similar symbols and logic to those used in fault tree analysis.
- c. MORT does not look at what happened during an accident, but traces causal factors back to management systems to identify why events happened.
- d. It is very effective in identifying the underlying management root causes of hazards.

Correct Answer: Select (c) **Explanation:** MORT has the capability to analyze what happened during an accident. It is a comprehensive, analytical, and disciplined method for determining the causes and contributing factors of an incident.

Question 17

Question: Pure risk is defined as risk that can only produce negative outcomes. Speculative risk is defined as risk that can produce both positive and negative outcomes. Which of the following is not an example of pure risk?

- a. Fire
- b. Unrecoverable disability
- c. Gambling
- d. Accident

Correct Answer: Select (c) **Explanation:** Gambling can offer winnings as well as losses.

Question 18

Question: You are a CSP in charge of safety for a petrochemical plant. A reactor explodes and 4 workers are killed and 3 are severely injured. The local newspapers and TV personnel pour into the plant and want information from you. Which of the following courses of actions would be most appropriate for you to take?

- a. Try to downplay the accident.
- b. Try to tell them everything that you know.
- c. Stay in the background and provide information to management.
- d. Try to blame the accident on unavoidable causes.

Correct Answer: Select (c) **Explanation:** In situations like this, the safety professional should stay in the background and provide adequate information to management (unless specifically directed by management to get directly involved).

Question 19

Question: As a safety and health professional in a midsize company, you have established a safety and health program. One of the employees in the production department has consistently been ignoring safety rules and policies. You have discussed the matter with his supervisor without getting any results. You should:

- a. Contact the company vice president
- b. Contact the personnel department
- c. Contact OSHA
- d. Contact the employee's labor union

Correct Answer: Select (b) **Explanation:** The best course of action is to inform the personnel department.

Question 20

Question: All of the following must be considered in developing a safety and health program for an organization EXCEPT:

- a. Emotional aspects
- b. Individual differences
- c. Motivational issues
- d. Availability of personal protective equipment

Correct Answer: Select (d) **Explanation:** PPE is not a consideration in developing a safety and health program. It is a consideration after the program has been developed.

Question 21

Question: Formal hazard analysis techniques can be divided into two general categories. Which of the following is the correct description of these two categories?

- a. HAZOP and Fault Tree
- b. FMEA and Event Tree
- c. FMEA and OHA
- d. Inductive and deductive

Correct Answer: Select (d) **Explanation:** Formal hazard analysis techniques are often divided into two general categories, inductive and deductive.

In inductive hazard analysis (bottom up), future behavior is based on experience which is hypothetical, and the conclusion is generally more than just the sum of observations. In deductive hazard analysis (top down), future behavior is concluded from a number of premises which may be true or false.

Question 22

Question: According to Berlo, the communication process is made up of seven steps. Which of the following is NOT one of the seven communication steps suggested by Berlo?

- a. Encoding
- b. The channel
- c. Interpretation
- d. Decoding

Correct Answer: Select (c) **Explanation:** According to Berlo, the communication process is made up of seven steps; the communication source, encoding, the message, the channel, decoding, the receiver and feedback.

Question 23

Question: Which of the following is NOT considered to be a valid approach in safety programs that primarily focus on behavior modification?

- a. Eliminate unwanted behavior.
- b. Provide training to eliminate or reduce the number of unsafe conditions.
- c. Educate employees on desired responses.
- d. Provide reward and/or praise for working safely.

Correct Answer: Select (b) **Explanation:** Training on eliminating unsafe conditions is not considered to be a behavioral modification approach to enhance safety programs.

Question 24

Question: All of the following are examples of situations that knowledge of the risks involved may not lead to safe behaviors EXCEPT:

- a. Smoking
- b. Handling toxic chemicals
- c. Eating high-cholesterol, high fat foods
- d. Not wearing seat belts

Correct Answer: Select (b) **Explanation:** Knowledge of the risks involved in handling toxic chemicals usually leads to safe behaviors.

Question 25

Question: All of the following are steps involved in conducting Job Hazard Analysis EXCEPT:

- a. Determine the steps involved in each job
- b. Identify hazards related to each task
- c. Quantify the results using minimum cut sets
- d. Determine how hazards can be mitigated

Correct Answer: Select (c) **Explanation:** Minimum cut sets are used in Fault Tree Analysis. A minimal cut set is the smallest combination of component failures which, if they all occur, will cause the top event to occur.

By definition, a minimal cut set is thus a combination (intersection) of primary events sufficient for the top event. The combination is a "smallest" combination, in that all the failures are needed for the top event to occur. If one of the failures in the cut set does not occur, then the top event will not occur (by this combination).

Any fault tree will consist of a finite number of minimal cut sets that are unique for that top event. The one-component minimal cut sets, if there are any, represent those single failures that will cause the top event to occur. The two-component minimal cut sets represent the double failures that together will cause the top event to occur.

Question 26

Question: Which of the following statements is true in regard to Root Cause Analysis?

- a. It combines the forward thinking pattern of PHA with the backward thinking pattern of FTA to predict the failure of safety functions that could lead to an undesirable event.
- b. It is the process by which the cause and effect relationship of an event, usually with undesirable consequences, are analyzed.
- c. It is a process by which the members of a team engage in brainstorming sessions to find out the deviation of process parameters from their design intent.
- d. It is best suited for predicting human errors by breaking a job into several steps and analyzing each step separately.

Correct Answer: Select (b) **Explanation:** Root Cause Analysis (RCA) is the process by which the cause and effect relationship of an event, usually with undesirable consequences, are analyzed. The goal of conducting an RCA is to find a solution that ensures that the undesirable event does not happen.

RCA uses a specific set of steps to find the main cause of a problem. This gives the analyst an opportunity to determine a) what happened, b) why it happened, and c) what to do in order to prevent its recurrence.

In RCA, the causes of a problem can be physical (for example, a control valve failed to open), human error (the technician failed to perform a scheduled maintenance on the control valve), and/or organizational causes (for example, no one was assigned to do maintenance and everyone thought someone else was assigned to do it).

Question 27

Question: Which of the following IS NOT considered an essential step in the process of risk management?

- a. Communication with supervisors
- b. Development of risk scenarios
- c. Means of control
- d. Implementation of best control method

Correct Answer: Select (a) **Explanation:** The fourth step in the process of risk management is management and control of the risk and feedback on the effectiveness of the process.

Question 28

Question: In the process of risk assessment, both the probability and severity of risk must be addressed. This can be done either graphically, with one axis showing the probability and the other axis showing the severity, or in the form of a table. This representation of risk data is called:

- a. Risk chart
- b. Risk matrix
- c. Risk table
- d. Risk domain

Correct Answer: Select (b) **Explanation:** The representation of risk data in graphical or tabular form is called a risk matrix. For example, a typical risk matrix plots Probability on one axis and Severity on the other.

Point A represents a relatively low probability and low severity risk. However, Point C has a high probability and high severity. Determining the acceptability or nonacceptability of risks is a management decision.

Question 29

Question: Which of the following techniques is most helpful to top management in placing priorities on corrective actions to eliminate or reduce risks?

- a. Fault tree analysis
- b. Risk analysis
- c. Failure analysis
- d. Cost/benefit analysis

Correct Answer: Select (d) **Explanation:** The benefit obtained from risk reduction must be justified through benefits to the organization.

Question 30

Question: All of the following are steps which must be undertaken in a risk assessment process EXCEPT:

- a. Hazard identification
- b. Decide on who and what may be harmed and how
- c. Evaluate the risks and decide on control measures
- d. The person in charge of the risk assessment process must attend risk assessment training

Correct Answer: Select (d) **Explanation:** The other steps involved in the risk assessment process are to record the findings and implement them, then review the assessment and update if necessary.

Question 31

Question: In proper risk assessment, a cost-benefit analysis must be performed. Both tangible and intangible costs must be considered. All of the following are examples of tangible costs associated with a project EXCEPT:

- a. Shipping
- b. Energy costs
- c. Real Estate
- d. Price of goods or equipment

Correct Answer: Select (b) **Explanation:** Energy costs are considered an intangible cost for a project. Tangible costs typically include things a business can buy directly for specific costs, such as labor, materials and space.

Question 32

Question: A measure of the combined probability and severity of potential harm to one or more resources as a consequence of exposure to one or more hazards is the correct definition of:

- a. Risk
- b. Risk assessment
- c. Risk control
- d. Risk management

Correct Answer: Select (a) **Explanation:** Risk is defined as the simple product of probability and severity. In order to reduce risk, the probability or severity or both must be reduced. For example, consider an exothermic (heat generating) reaction in a chemical reactor where the heat of reaction is removed by pumping a coolant through the jacket around the reactor. If the coolant pump fails, heat will accumulate in the reactor leading to a runaway reaction and explosion. Installation of a spare pump would reduce the probability of losing coolant flow to the reactor because two pumps must fail at the same time.

Question 33

Question: The process of determining the degree of threat that is posed by one or more hazards is called:

- a. Risk control
- b. Risk management
- c. Risk assessment
- d. Risk acceptance

Correct Answer: Select (c) **Explanation:** Risk may be assessed either subjectively (non-quantitatively) or numerically (quantitatively).

Question 34

Question: The process of minimizing or eliminating accidental losses by anticipating and preventing the occurrence of unplanned events is called?

- a. Risk assessment
- b. Risk management
- c. Risk acceptance
- d. Risk control

Correct Answer: Select (d) **Explanation:** Risk control incorporates a wide range of elements including worker safety. Once the hazards are identified and the risk assessed, risk should be managed and controlled. Engineering and administrative measures are used to control risk. For example, use of enclosures to protect workers is an example of engineering control and measures such as job rotation is an example of administrative control.

Question 35

Question: The professional assessment of all loss potentials in an organization's structure and operations leading to the establishment and administration of a comprehensive loss control program is called:

- a. Risk control
- b. Risk assessment
- c. Risk management
- d. Risk acceptance

Correct Answer: Select (c) **Explanation:** Risk management's goal is to eliminate or to reduce losses to an acceptable minimum at the lowest possible cost.

Question 36

Question: "Risk homeostasis" is a theory that:

- a. A person may operate at a given risk level regardless of the safeguards provided
- b. Deals with non-industrial risks such as smoking
- c. Deals with risk of hazards encountered off the job such as climbing a ladder at home
- d. All of the above

Correct Answer: Select (a) **Explanation:** For example, a worker who feels he/she is protected by a back-belt may attempt to lift even more with the belt than he/she would without it.

Question 37

Question: Fault Tree Analysis (FTA) is:

- a. Bottom up system safety technique
- b. Top down system safety technique
- c. Used to determine the deviation in process parameters from their design intent
- d. A qualitative system safety technique that identifies the consequences of success or failure of safety functions in a system.

Correct Answer: Select (a) **Explanation:** Fault tree starts with an undesirable event and traces backward to all possible causes. Failure mode effect analysis (FMEA) is also a bottom up system safety technique. Event Tree Analysis (ETA) is a system safety technique that determines the consequences of success or failure of safety functions in a system.

Question 38

Question: Which of the following BEST describes the concept of safety and health "benchmarking"?

- a. Benchmarking cannot be applied to ALL aspects of a safety and health program.
- b. Benchmarking does not necessarily require management commitment but it does require employee involvement.
- c. Any information that is exchanged does not have to be comparable with partner's information.
- d. Safety and health benchmarking compares an organization's safety and health performance with similar organizations.

Correct Answer: Select (d) **Explanation:** Benchmarking requires both management commitment and employee involvement. Information that is exchanged must be comparable to the other organization's information and arrangements for confidentiality issues must be made.

Question 39

Question: In performing a Job Safety Analysis (JSA), several factors affect prioritizing different jobs for analysis. All of the following are factors which affect the prioritization of jobs for analysis EXCEPT:

- a. Incident frequency
- b. Incident severity
- c. New jobs
- d. Complexity of an operation

Correct Answer: Select (d) **Explanation:** Complexity of an operation is not considered to be a factor in prioritizing a job for Job Safety Analysis.

Question 40

Question: All the following statements are correct about Job Safety Analysis (JSA) EXCEPT:

- a. JSA breakdowns a job into several steps.
- b. JSA identifies the hazards associated with each step.
- c. JSA can be used to educate employees on safe practices.
- d. JSA can quantify the accident probabilities.

Correct Answer: Select (d) **Explanation:** JSA breaks down a job into many steps and identifies the hazards associated with each step and their means of control. This technique can be used to educate employees on safe work practices. It must be pointed out that the priority for JSA should be assigned to those jobs that have a relatively high probability of leading to an accident which has severe consequences.

Question 41

Question: In conducting a Job Safety Analysis, the priority should go to which of the following jobs?

- a. Jobs with the highest injury or illness rates.
- b. Jobs with the potential to cause severe or disabling injuries or illnesses.
- c. Jobs in which one simple human error could lead to a severe accident or injury.
- d. All the above.

Correct Answer: Select (d) **Explanation:** A job hazard analysis can be conducted on many jobs in the workplace. Priority should go to the following types of jobs:

- Jobs with the highest injury or illness rates.
- Jobs with the potential to cause severe or disabling injuries or illnesses, even if there is no history of previous accidents.
- Jobs in which one simple human error could lead to a severe accident or injury.
- Jobs that are new to the facility operation or have undergone changes in processes and procedures.
- Jobs complex enough to require written instructions.

Question 42

Question: Which of the following best describes the "Casual Analysis?"

- a. To find the root cause of a problem instead of finding the symptoms.
- b. Helps to uncover the facts that lead to a certain situation.
- c. Cause refers to a reason.
- d. All the above.

Correct Answer: Select (d) **Explanation:** Cause refers to a reason. The purpose of causal analysis is trying to find the root cause of a problem instead of finding the symptoms. This technique helps to uncover the facts that lead to a certain situation.